

B.Sc. Semester III (Honours) Examination, 2018-19**MATHEMATICS****Course ID : 32114****Course Code : SHMTH-304GE-3(T)****Course Title : Algebra****Time: 2 Hours****Full Marks: 40***The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.***1. Answer any five questions:****2×5=10**

- (a) Find the values of $i^{3/4}$.
- (b) Apply Descartes' rule of signs to find the nature of the roots of the equation $x^4 + 7x^3 + 2x - 1 = 0$.
- (c) If a, b, c are positive reals, not all equal, then prove that $(a + b + c)(bc + ca + ab) > 9abc$.
- (d) Let S be the set of all lines in 3-space. A relation R is defined on S by " lRm if and only if l lies on the plane of m " for $l, m \in S$. Examine whether R is an equivalence relation.
- (e) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 3x^2 - 5$. Then find $f^{-1}(\{70\})$.
- (f) Use Cayley-Hamilton theorem to find A^{-1} where $A = \begin{pmatrix} 3 & -2 \\ 4 & -3 \end{pmatrix}$.
- (g) Find the dimension of the subspace S of $\mathbb{R}^4$, where $S = \{(x, y, z, w) \in \mathbb{R}^4 : x + 2y - z + 2w = 0\}$.
- (h) Examine whether the mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x + y, y + z, x + 1)$ is linear or not? Justify your answer.

2. Answer any four questions:**5×4=20**

- (a) Solve: $x^3 - 9x + 28 = 0$. 5
- (b) (i) If the roots of the equation $x^3 + ax^2 + bx + c = 0$ be in G.P, show that $b^3 = a^3c$.
- (ii) Show that $3x^5 - 4x^2 + 6 = 0$ has at least two imaginary roots. 3+2=5
- (c) (i) If x, y, z are positive real numbers and $x + y + z = 1$, prove that $8xyz \leq (1 - x)(1 - y)(1 - z) \leq \frac{8}{27}$.
- (ii) If n be a positive integer >1 , prove that $\left(\frac{n+1}{2}\right)^n > n!$ 3+2=5

(d) The matrix of $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ relative to the ordered basis $\{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}$ is

$$\begin{pmatrix} 0 & 3 & 0 \\ 2 & 3 & -2 \\ 2 & -1 & 2 \end{pmatrix}. \text{ Find the linear transformation } T. \text{ Find the matrix of } T \text{ relative to the ordered}$$

basis $\{(2, 1, 1), (1, 2, 1), (1, 1, 2)\}$.

3+2=5

(e) (i) Use the theory of Congruences to prove that $17|(2^{3n+1} + 3 \cdot 5^{2n+1})$ for all positive integers n .

(ii) Let a, b, c, m be positive integers such that $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$ then prove that $a \equiv b \pmod{m}$.

3+2=5

(f) Solve if possible, the system of equations

5

$$x_1 + 2x_2 - x_3 = 0$$

$$-x_1 + x_2 + 2x_3 = 2$$

$$2x_1 + x_2 - 3x_3 = 2$$

3. Answer any one question:

10×1=10

(a) (i) If α, β are the roots of the equation $x^2 - 2x + 4 = 0$, prove that

$$\alpha^n + \beta^n = 2^{n+1} \cos \frac{n\pi}{3}.$$

(ii) Show that the sum of the 99th power of the roots of the equation $x^5 = 1$ is zero.

(iii) Verify Cayley-Hamilton theorem for the matrix $\begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$. Hence find A^{-1} .

3+3+4=10

(b) (i) Apply elementary row operations to reduce the following matrix to a row echelon

$$\text{matrix and find the rank of the matrix } \begin{pmatrix} 2 & 0 & 4 & 2 \\ 3 & 2 & 6 & 5 \\ 5 & 2 & 10 & 7 \\ 0 & 3 & 2 & 5 \end{pmatrix}.$$

(ii) State first principle of mathematical induction. Apply this to prove that $3^{2n} = 8n - 1$ is divisible by 64.

(4+1)+(2+3)=10

B.Sc. 3rd Semester (Honours) Examination, 2019-20**MATHEMATICS****Course ID : 32114****Course Code : SHMTH-304/GE-3****Course Title: Algebra****Time: 2 Hours****Full Marks: 40***The figures in the right hand side margin indicate marks.**Candidates are required to give their answers in their own words as far as practicable.**Unless otherwise mentioned, notations and symbols have their usual meaning.***1. Answer any five questions:****2×5=10**

- (a) If $Z + \frac{1}{Z} = \sqrt{3}$, then show that $Z^6 = -1$.
- (b) Show that $n^n \geq n!$ for all $n \in \mathbb{N}$.
- (c) Give an example of transitive relation which is
 - (i) symmetric but not reflexive and
 - (ii) reflexive but not symmetric.
- (d) Transform the equation to remove the square from $x^3 - 15x^2 - 33x + 84 = 0$.
- (e) Apply Descartes rule of signs to examine the nature of the roots of the equation $x^4 + 2x^2 + 3x - 1 = 0$.
- (f) Use division algorithm, find integers u and v satisfying $63u + 55v = 1$
- (g) Find the dimension of the subspace W of $\mathbb{R}^4$, where $W = \{(x, y, z, w) \in \mathbb{R}^4 : x + 2y - z = 0, 2x + y + w = 0\}$
- (h) Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(x, y) = (2x + y, y)$, $(x, y) \in \mathbb{R}^2$ Examine whether T is linear or not? Justify your answer.

2. Answer any four questions:**5×4=20**

- (a) (i) Show that the product of all values of $(1 + i\sqrt{3})^{\frac{3}{4}}$ is 8.
- (ii) State Cauchy-Schwarz inequality. 3+2=5
- (b) Define linear transformation. Find out the matrix corresponding to the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ s. t.

$$T(x, y, z) = \left(2x + 2y + z, \frac{-x + y + 3z}{2} \right)$$

with the respect to the ordered basis $\{(0, 1, 1)(1, 0, 1)(1, 1, 0)\}$ of $\mathbb{R}^3$ and $\{(1, 0), (1, 1)\}$ of $\mathbb{R}^2$.

(c) Find the eigenvalues and eigenvectors of the matrix:

$$\begin{pmatrix} 1 & 1 & 1 \\ -1 & -1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

(d) (i) Find the product of all values of $(1 + i\sqrt{3})^{\frac{3}{4}}$

(ii) If $x + \frac{1}{x} = 2 \cos \theta$ and θ is real, prove that $x^n + \frac{1}{x^n} = 2 \cos n\theta$. 3+2=5

(e) (i) Prove that the product of any m consecutive integers is divisible by m .

(ii) If a is prime to b , prove that a^2 is prime to b and a^2 is prime to b^2 . 3+2=5

(f) Solve, if possible, the system of equations

$$x_1 + 2x_2 - x_3 = 10$$

$$-x_1 + x_2 + 2x_3 = 2$$

$$2x_1 + x_2 - 3x_3 = 8$$

3. Answer any one question:

10×1=10

(a) (i) Find the remainder when $1! + 2! + \dots + 50!$ is divided by 15.

(ii) If Z is a non-zero complex number and p, q, m and n are positive integers, where $\frac{p}{q} = \frac{m}{n}$ with $\gcd(m, n) = 1$, then $Z^{\frac{p}{q}} = Z^{\frac{m}{n}}$.

(iii) Solve the equation using Cardan's method $x^3 - 18x - 35 = 0$ 2+3+5=10

(b) (i) State Cayley-Hamilton theorem. Verify Cayley-Hamilton theorem for the matrix

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 1 \\ 2 & 3 & 2 \end{pmatrix} \text{ and hence compute } A^{-1}.$$

(ii) The matrix of a linear mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ relative to the ordered bases $\{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}$ of $\mathbb{R}^3$ and $\{(1, 0), (1, 1)\}$ of $\mathbb{R}^2$ is $\begin{pmatrix} 1 & 2 & 4 \\ 2 & 1 & 0 \end{pmatrix}$. Find the

linear transformation T . Find the matrix of T relative to the ordered bases

$\{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ of $\mathbb{R}^3$ and $\{(1, 1), (0, 1)\}$ of $\mathbb{R}^2$. (1+3+1)+(3+2)=10

B.SC. THIRD SEMESTER (HONS.) EXAMINATION 2021

Subject: Mathematics

Course ID: 32114

Course Title: Algebra

Course Code: SH/MTH/304/GE-3

Full Marks: 40

Time: 2 hour

The figures in the margin indicate full marks

Notations and symbols have their usual meaning.

1. Answer *any five* of the following questions: (2x5=10)

- a) Prove that $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n.(n+1)} = \frac{n}{n+1}$.
- b) Prove that $\frac{(n+1)^n}{2^n} > n!$.
- c) Show that the product of all values of $(\sqrt{3} + i)^{\frac{3}{5}}$ is $8i$.
- d) Apply Descartes' rule of sign to examine the nature of roots of the equation
$$x^4 - 4x^3 + 2x^2 + 4x + 1 = 0.$$
- e) What is the smallest equivalence relation on the set $A = \{a, b, c\}$.
- f) Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = |x| + x$ is neither injective nor surjective.
- g) Find the dimension of the subspace W of $\mathbb{R}^3$ defined by
$$W = \{(x, y, z) \in \mathbb{R}^3 : 2x + y - z = 0\}.$$

- h) Use Cayley-Hamilton theorem to compute A^{2020} , where $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

2. Answer *any four* of the following questions: (5x4=20)

- a) If $(1+x)^n = a_0 + a_1x + a_2x^2 + \dots$, prove that
$$a_0 + a_4 + a_8 + \dots = 2^{n-2} + 2^{\frac{n-2}{2}} \cos \frac{n\pi}{4}.$$
- b) If one root of the equation $x^3 + ax + b = 0$ is twice the difference of the other two, then prove that one root is $\frac{13b}{3a}$.
- c) (i) Show that $1! . 3! . 5! \dots (2n-1)! > (n!)^n$.
(ii) Prove that the number of primes is infinite. 3 + 2
- d) Determine the conditions of a and b for which the system of equations has (i) only one solution, (ii) no solution, (iii) many solutions:

$$\begin{aligned} x + 2y + z &= 1 \\ 2x + y + 3z &= b \\ x + ay + 3z &= 1 + b. \end{aligned}$$

e) A mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is defined by

$$T(x, y, z) = (x + y + z, 2x + y + 2z, x + 2y + z), (x, y, z) \in \mathbb{R}^3.$$

Show that T is a linear mapping. Find $\text{Ker } T$ and the dimension of $\text{Ker } T$.

f) Find the real eigen values (if any) and the corresponding eigen vectors of the matrix

$$\begin{bmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \\ 3 & -3 & 6 \end{bmatrix}.$$

3. Answer *any one* of the following questions:

(1x10=10)

a) (i) Let z be a variable complex number such that the amplitude of $\frac{z-i}{z+1}$ is $\frac{\pi}{4}$. Show that the point z lies on a circle in the complex plane.

(ii) Prove that the product of any m consecutive integers is divisible by m .

(iii) Show that the mapping $f: S \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{1-|x|}$, where $S = (-1, 1)$, is

bijjective.

4+3+3

b) (i) Prove that the eigen values of a real symmetric matrix are all real.

(ii) Determine the linear mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that maps the basis vectors

$(0,1,1), (1,0,1), (1,1,0)$ of $\mathbb{R}^3$ to the vectors $(2,1,1), (1,2,1), (1,1,2)$ of $\mathbb{R}^3$ respectively. Also

find $\text{Ker } T$ and $\text{Im } T$.

5+5

B.SC. THIRD SEMESTER (HONOURS) EXAMINATIONS, 2022

Subject: Mathematics

Course ID: 32114

Course Code: SH/MTH/304/GE-3

Course Title: Algebra

Full Marks: 40

Time: 2 Hours

The figures in the margin indicate full marks

Notations and symbols have their usual meaning.

1. Answer *any five* of the following questions:

2x5=10

a) If z is any Complex number where $z = x + iy$, then prove that

$$|x| + |y| \leq \sqrt{2} |z|.$$

b) Prove that $2^n > 1 + \sqrt{2^{n-1}} n$.

c) Using mathematical induction prove that there are 2^n subsets of a set of n elements.

d) Find the equation whose roots are the squares of the roots of the equation

$$x^4 - x^3 + 2x^2 - x + 1 = 0.$$

e) How many symmetric relations can be defined on a set with n elements?

f) Two sets A and B contain 5 and 7 elements respectively. Find the total number of mappings from A to B . How many of them are injective?

g) If A is a non-singular matrix of order 4, then determine the rank of the matrix A^4 .

h) Find all the real values of x for which the rank of the matrix A is 2, where

$$A = \begin{pmatrix} 1 & 2 & 3 & 1 \\ 2 & 5 & 3 & x \\ 1 & 1 & 6 & x+1 \end{pmatrix}.$$

2. Answer *any four* of the following questions:

5x4=20

a) Using De Moivre's theorem prove that

$$\sin^4 \theta \cos^2 \theta = \frac{1}{32} [\cos 6\theta - 2 \cos 4\theta - \cos 2\theta + 2].$$

b) If a, b, c are three positive numbers and $a + b + c = 1$, then prove that,

$$\left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 + \left(c + \frac{1}{c}\right)^2 \geq 33\frac{1}{3}$$

c) Let $\alpha, \beta, \gamma, \delta$ be the roots of the equation, $x^4 + px^3 + qx^2 + rx + s = 0$,

such that $\alpha\beta + \gamma\delta = 0$. Then prove that $p^2s + r^2 - 4qs = 0$.

d) Determine the conditions for which the system of equations has (a) only one solution, (b) no solution, (c) many solutions

$$x + y + z = b, \quad 2x + y + 3z = b + 1, \quad 5x + 2y + az = b^2.$$

e) If $U = L\{(2,0,1), (3,1,0)\}$ and $W = L\{(2,0,1), (3,1,0)\}$, then find $\dim(U + W)$ and $\dim(U \cap W)$.

f) Find the eigen values and the corresponding eigen vectors of the real matrix

$$\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}.$$

3. Answer *any one* of the following questions:

10x1=10

(a) (i) Find the locus of the point z given by, $\arg \left\{ \frac{z-1}{z+2} \right\} = \pm \frac{\pi}{2}$.

(ii) Find the greatest number K for which $5x + 7y = K$ has exactly 9 solutions in non-negative integers.

(iii) Correct or justify the statement: 'Every relation is a mapping but every mapping is not a relation'.

4+4+2=10

(b) (i) Solve the system of equations

$$x + 2y + z = 1, \quad 3x + y + 2z = 3, \quad x + 7y + 2z = 1.$$

(ii) If x, y, z are three positive numbers, such that $x^2 + y^2 + z^2 = 27$, then prove that

$$x^3 + y^3 + z^3 \geq 81.$$

(ii) If α, β, γ are the roots of the equation $x^3 + 3x^2 + 10 = 0$, then prove that

$$\alpha^4 + \beta^4 + \gamma^4 = 201.$$

$$4+3+3=10$$

B.Sc. 2nd Semester (G.E.), Internal Exam, 2022-23

Subject: Mathematics Course Title: Algebra

F.M = 10 marks.

Time: 30 min.

Answer any two questions:

$$5 \times 2 = 10$$

(1) If z be a complex number and $\frac{z+1}{z-i}$ be purely imaginary, then show that z lies on the circle whose centre is at $\frac{1}{2}(-1+i)$ and the radius is $\frac{1}{\sqrt{2}}$.

(2) State the principle of Mathematical Induction and using it to show that $(5^n - 1)$ is divisible by 4 for all positive integers n .

(3) Find the Echelon form of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 4 & 6 & 8 \end{bmatrix}$

and hence find the rank of A .

(4) If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be such that $T(a, b) = (a, 0)$ then show that T is a linear transformation.

B.Sc. Semester-II Examination, 2022-23**MATHEMATICS [Honours]****Course ID : 22114****Course Code : SH/MTH/203/GE-2****Course Title : Algebra****[NEW SYLLABUS]****Time : 2 Hours****Full Marks : 40***The figures in the right-hand margin indicate marks.**Candidates are required to give their answers in their own words as far as practicable.**Notations and symbols have their usual meaning.***UNIT-I**

1. Answer any **five** from the following questions:

 $2 \times 5 = 10$

- a) Find the number of positive and negative roots of the equation $x^5 - x^4 + x^3 + 8x^2 + 2x - 2 = 0$.
- b) Prove that if $x^2 = e$ for all element of a group G , then G is a commutative.
- c) Z is a Variable complex number such that an amplitude of $\frac{Z-i}{Z+1}$ is $\frac{\pi}{4}$. Show that the point Z lies on a circle in the complex plane.

[Turn Over]

- d) Show that for any three real number a, b, c , $a^8 + b^8 + c^8 \geq a^2 b^2 c^2 (ab + bc + ca)$.
- e) Let W be the subspace of $\mathbb{R}^3$ defined by
 $W = \{(x, y, z) \in \mathbb{R}^3 : 2x + 2y + z = 0,$
 $3x + 3y - z = 0, x + y - 3z = 0\}$. Find the dimension of W .
- f) If a is prime to b , prove that a^2 is prime to b and b^2 .
- g) If α, β and γ be the roots of the equation $x^3 + 2x^2 - 3x - 1 = 0$. Find the value of $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha}$.
- h) The eigen values of the matrix A are 5 and 3. Find the eigen values of the matrix $A^2 - 8A + 15I + 30A^{-1}$.

UNIT-II

2. Answer any **four** from the following questions:

$$5 \times 4 = 20$$

- a) i) Let $\rho = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid 3a + 4b = 7n, \text{ for some } n \in \mathbb{Z}\}$. Show that ρ is an equivalence relation.
- ii) Use Principle of mathematical induction to prove that $2 \cdot 7^n + 3 \cdot 5^n - 5$ is divisible by 24 for all $n \in \mathbb{N}$.

$$3 + 2 = 5$$

- b) i) Let $a_i > -\frac{1}{3}$, ($i = 1, 2, 3$) and $a + b + c = 1$.
Apply Cauchy-Schwarz inequality to prove that $\sqrt{3a+1} + \sqrt{3b+1} + \sqrt{3c+1} \leq 3\sqrt{2}$.
- ii) Find the minimum value of $3x + 2y$ where x, y are positive real numbers satisfying the condition $x^2 y^3 = 48$. $3+2=5$
- c) If α, β, γ be the roots of the equation $x^3 - px^2 + r = 0$, find the equation whose roots are $\frac{1}{\alpha^2} + \frac{1}{\beta^2} - \frac{1}{\gamma^2}, \frac{1}{\beta^2} + \frac{1}{\gamma^2} - \frac{1}{\alpha^2}, \frac{1}{\gamma^2} + \frac{1}{\alpha^2} - \frac{1}{\beta^2}$.
- d) If H and K are subgroups of the group $(G, 0)$ then HK is subgroup of G , iff $HK = KH$.
- e) Reduce the reciprocal equation $x^5 - 6x^4 + 7x^3 + 7x^2 - 6x + 1 = 0$ to its standard form and solve it.
- f) Find the linear mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ which maps the basis vectors $(0, 1, 1), (1, 0, 1), (1, 1, 0)$ of $\mathbb{R}^3$ to the vectors $(2, 0, 0), (0, 2, 0), (0, 0, 2)$ of $\mathbb{R}^3$ respectively. Find $\text{Ker } T$ and $\text{Im } T$. Show that $\dim \text{Ker } T + \dim \text{Im } T = \dim \mathbb{R}^3$. $3+2=5$

UNIT-III

3. Answer any one of the following questions:

10×1=10

a) i) Let $f: \mathbb{R} \rightarrow S = \{x \in \mathbb{R} : 1 < x < 1\}$ defined by $f(x) = \frac{x}{1+|x|}$, $x \in \mathbb{R}$. Find f^{-1} .

ii) Apply Descartes rule of signs to ascertain the minimum number of complex roots of the equation $x^6 - 3x^2 - 2x - 3 = 0$.

iii) Find a matrix P and a diagonal matrix D such that $P^{-1}AP = D$, where $A = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix}$.

3+2+5=10

b) i) If α, β, γ be the roots of the equation $x^3 + px^2 + qx + r = 0$, find the value of $\sum \alpha^3 \beta^3$.

ii) Show that $\text{Log}(1+i)^3 \neq 3\text{Log}(1+i)$.

iii) Show that the ratio of the principal values of $(1+i)^{1-i}$ and $(1-i)^{1+i}$ is $\sin(\log 2) + i \cos(\log 2)$.

4+2+4=10

Q

B.Sc. Semester-III Examination, 2022-23**MATHEMATICS [Honours]****Course ID : 32114****Course Code : SH/MTH/304/GE-3****Course Title : Algebra****Time : 2 Hours****Full Marks : 40***The figures in the right-hand margin indicate marks.**Candidates are required to give their answers in their own words as far as practicable.**Notations and symbols have their usual meaning.***UNIT-I****1. Answer any five of the following questions:** $2 \times 5 = 10$ **a) Prove that for any complex number**

$$z, |z| \geq \frac{1}{\sqrt{2}} (|\operatorname{Re} z| + |\operatorname{Im} z|).$$

b) Construct an equivalence relation on the set $A = \{1, 2, 3\}$.**c) Using principle of mathematical induction prove that $3^{2n} - 8n - 1$ is divisible by 64 where n is a positive integer.****d) Apply Descartes rule of signs to examine the nature of the roots of the equation $x^5 + x^3 - x^2 = 0$.***[Turn Over]*

- e) Let λ be an eigenvalue of an $n \times n$ matrix A . Show that λ^4 is an eigenvalue of the matrix A^4 .
- f) Show that the mapping $f: S \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{1-|x|}$, where $S = (-1, 1)$, is bijective.
- g) Prove that $S \times S$ is an equivalence in S .
- h) Express the matrix A as a product of elementary matrices, where $A = \begin{pmatrix} 3 & 2 \\ 2 & 5 \end{pmatrix}$.

UNIT-II

2. Answer any **four** of the following questions:

$$5 \times 4 = 20$$

- a) Show that one of the values of

$$\left(1+i\sqrt{3}\right)^{\frac{3}{4}} + \left(1-i\sqrt{3}\right)^{\frac{3}{4}} \text{ is } \sqrt[4]{32}.$$

- b) Let $T: R^2 \rightarrow R^3$ defined by

$$T(x, y) = (x + y, x - 2y, 3x + y).$$

Show that T is non-singular and find T^{-1} .

- c) If α, β, γ are the roots of the equation $x^3 + qx + r = 0$, then find the equation whose roots are $\frac{\beta + \gamma}{\alpha^2}, \frac{\gamma + \alpha}{\beta^2}, \frac{\alpha + \beta}{\gamma^2}$.

d) A mapping $T: R^3 \rightarrow R^3$ is defined by

$$T(x, y, z) = (x + 2y + 3z, 3x + 2y + z, x + y + z), (x, y, z) \in R^3.$$

Show that T is a linear mapping. Find $\text{Ker } T$ and the dimension of $\text{Ker } T$.

e) Show that the solutions of the equation

$$(1+x)^{2n} + (1-x)^{2n} = 0 \quad \text{are} \quad x = \pm i \tan \frac{(2r-1)\pi}{4n},$$

$$r = 1, 2, \dots, n.$$

f) i) If $d = \gcd(a, m)$, then show that

$$ax \equiv ay \pmod{m} \leftrightarrow x \equiv y \pmod{\frac{m}{d}}.$$

ii) Find the least positive residues in

$$3^{36} \pmod{77}.$$

$$3+2=5$$

UNIT-III

3. Answer any one of the following questions:

$$10 \times 1 = 10$$

a) i) Find the sum of 99th powers of the roots of the equation $x^7 - 1 = 0$.

ii) Show that $3x^5 - 4x^2 + 6 = 0$ has at least two imaginary roots.

iii) Diagonalise the matrix

$$A = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}. \quad 3+2+5=10$$

b) i) Find the linear mapping $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ if $T(1, 0, 0) = (2, 3, 4)$, $T(0, 1, 0) = (1, 2, 3)$, $T(0, 0, 1) = (1, 1, 1)$. Find the matrix of T relative to the ordered basis $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.

ii) Find the dimension of the subspace W of $\mathbb{R}^3$ where

$$W = \{(x, y, z) : x + 2y + z = 0, 2x + y + 3z = 0\}.$$

iii) If p is prime, greater than 3, show that 24 divides $(p^2 - 1)$. $(2+2)+3+3=10$

B. Sc. Semester II Examinations, 2023-24

Subject: Mathematics

Course ID: 22114

Course Code: SH/MTH/203/GE-2

Course Title: Algebra (NEW CBCS Syllabus 2022-25)

Time: 2 Hours.

Full Marks: 40

*The figures in the right-hand margin indicate marks.
Candidates are required to give their answer in their own words as far as practicable.*

Notations and Symbols have their usual meaning.

1. Answer any 5 (five) of the following questions:

(2 × 5 = 10)

a) If α, β, γ be the roots of the equation $x^3 + qx + r = 0$ then find the value of $\sum \frac{\alpha}{\beta}$.

b)) Find the value of $(\sqrt{3} - i)^{\frac{1}{7}}$.

c) If $S = a + b + c$, prove that $\frac{s}{s-a} + \frac{s}{s-b} + \frac{s}{s-c} > \frac{9}{2}$ where a, b, c are unequal positive real numbers.

d) Show that the mapping $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x, y) = (e^x, e^y)$ is not linear.

e) Find the rank of the matrix $\begin{pmatrix} 1 & 3 & 5 & -1 \\ 2 & 1 & -2 & 8 \\ 0 & 5 & 12 & -10 \end{pmatrix}$.

[Turn Over]

f) If a/b and a/c then prove that $a/(bx + cy)$ for any integers x and y .

g) Let $A = \{1, 2, 3\}$. Verify whether the relation $R = \{(1, 1), (2, 2), (3, 3), (2, 3), (3, 2), (1, 2), (2, 1)\}$ is transitive.

h) If $f: R \rightarrow R$ and $g: R \rightarrow R$ be two mappings, then compute $f \circ g$ with usual notations where

$$f(x) = x^2 + 3x + 2 \text{ and } g(x) = 3x - 1 \quad \forall x \in R.$$

UNIT II

2. Answer any 4 (four) of the following questions:

(5 × 4 = 20)

- a) State De Moivre's theorem. Prove it for integral values of the index. 1 + 4
- b) If α, β, γ are the roots of the equation $2x^3 + 3x^2 - x - 1 = 0$, then obtain the equation whose roots are $\frac{\alpha}{\beta+\gamma}, \frac{\beta}{\gamma+\alpha}, \frac{\gamma}{\alpha+\beta}$. Hence find the value of $\sum \frac{\alpha}{\beta+\gamma}$. 4 + 1
- c) Prove that the mapping $f: A \rightarrow B$ is invertible if and only if it is bijective. 5
- d) i) Express the vector $(1, 7, -4)$ as a linear combination of the vectors $(1, -3, 2)$ and $(2, -1, 1)$ in the vector space R^3 .
- ii) For what value of c the set of vectors $\{(1, 1, 2), (1, c, 3), (2, 0, c)\}$ is linearly independent. 3 + 2
- e) Obtain the Sturm's function of the equation $x^3 + 11x^2 - 102x + 181 = 0$. 5

f) i) State the Descartes' rule of signs. Also find the nature of the roots of $x^4 + 15x^2 + 7x - 11 = 0$.

ii) Prove that $3^{2n-1} + 2^{n+1}$ is divisible by 7 $\forall n \in \mathbb{N}$. $2 + 3$

UNIT III

3. Answer any 1 (one) of the following questions:

(10 × 1 = 10)

a) (i) If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then prove that $a + c \equiv b + d \pmod{m}$.

(ii) Find the rank of the matrix $\begin{pmatrix} 3 & 1 & 2 \\ 6 & 2 & 4 \\ 3 & 4 & 5 \end{pmatrix}$.

(iii) Determine the conditions for which the system of equations

$$\begin{aligned} x + y + z &= 6 \\ x + 2y + 3z &= 10 \\ x + 2y + az &= b \end{aligned}$$

has no solution, unique solution & infinite no. of solutions.

$2 + 2 + 6$

b) (i) Find the dimension of the subspace of S of $\mathbb{R}^4$, where

$$S = \{(x, y, z, w) \in \mathbb{R}^4 : x + 2y - z + 2w = 0\}.$$

(ii) Verify Cayley-Hamilton theorem for the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$.

(iii) Solve the cubic equation $x^3 - 6x - 9 = 0$ by Cardan's method.

$3 + 2 + 5$

B.Sc. 3rd Semester (Honours) Examination, 2023-24

MATHEMATICS

Course ID : 32114 Course Code : SH-MTH-304/GE-3

Course Title : Algebra

[Syllabus - 2017]

Time : 2 Hours

Full Marks : 40

The figures in the right margin indicate marks.

Candidates are required to answer in their own words as far as practicable.

Symbols and notations have their usual meaning.

UNIT-I

1. Answer **any five** from the following questions:

$$5 \times 2 = 10$$

- a) Find the values of $(-i)^{\frac{2}{3}}$.
- b) Apply Descartes' rule of sign to examine the nature of the roots of the equation
$$x^6 + 4x^4 + 2x^2 + 4x + 1 = 0.$$
- c) Construct an equivalence relation on the set $A = \{a, b, c\}$.
- d) Show that the function $f : R \rightarrow R$ defined by $f(x) = |x| + x$ is neither injective nor surjective.
- e) For any non empty set S , prove that $S \times S$ is an equivalence in S .

- f) Show that W is a subspace of $\mathbb{R}^4$ defined by
 $W = \{(x, y, z, w) : x, y, w \in \mathbb{R}, 3x + y + z + 2w = 0\}$.
- g) Let λ be an eigen value of an idempotent matrix A . Show that λ is 1 or 0.
- h) Find the rank of the matrix A^2 , where

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

UNIT-II

2. Answer *any four* from the following questions:

$$4 \times 5 = 20$$

- a) If
 $\cos \alpha + \cos \beta + \cos \gamma = 0, \sin \alpha + \sin \beta + \sin \gamma = 0$
 then prove that
 (i) $\sin(\alpha + \beta) + \sin(\gamma + \beta) + \sin(\gamma + \alpha) = 0$
 (ii) $\sin 3\alpha + \sin 3\beta + \sin 3\gamma = 3 \sin(\alpha + \beta + \gamma)$.
- $$3 + 2 = 5$$
- b) i) Show that $1! \cdot 3! \cdot 5! \dots (2n-1)! > (n!)^n$.
 ii) Prove that number of primes is infinite.
- $$3 + 2 = 5$$
- c) For what values of a the following system of equations is consistent? Solve in that case.

$$\begin{aligned}x - y + z &= 1 \\x + 2y + 4z &= a \\x + 4y + 6z &= a^2\end{aligned}$$

5

- d) Solve by Cardan's method $x^3 + 3x^2 - 3 = 0$:
- e) Use Cayley-Hamilton theorem to compute A^{2023} where

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

5

- f) The matrix of $T : R^3 \rightarrow R^3$ relative to the order basis $\{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}$ is

$$\begin{pmatrix} 0 & 3 & 0 \\ 2 & 3 & -2 \\ 2 & -1 & 2 \end{pmatrix}$$

Find the linear transformation

T . Find the matrix of T relative to the ordered basis $\{(2, 1, 1), (1, 2, 1), (1, 1, 2)\}$. 3+2

UNIT-III

3. Answer *any one* from the following questions:

$$1 \times 10 = 10$$

- a) i) Find the least positive residue in $2^{41} \pmod{23}$.
- ii) A mapping $T : R^3 \rightarrow R^3$ is defined by

$$T(x, y, z) = (x + y + z, 2x + y + 2z, x + 2y + z), (x, y, z) \in \mathbb{R}^3.$$

Show that T is a linear mapping. Find $\text{Ker } T$ and the dimension of $\text{ker } T$.

$$5+5=10$$

b) i) If $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$, find all eigenvalues of

A and obtain the all eigenvectors corresponding to all eigenvalues.

ii) If α, β, γ are the roots of the equation $x^3 + 3x^2 + 10 = 0$, then prove that $\alpha^4 + \beta^4 + \gamma^4 = 201$.

iii) Show that $3x^5 - 4x^2 + 6 = 0$ has at least two imaginary roots. 5+3+2